

Joe Kuwano

Snells Beach, Auckland

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Personal Summary

I am a second-year mechanical engineering student at the University of Auckland with a strong foundation in problem-solving, teamwork, and technical skills with experience in mechanical design, manufacture, electronics and hardware.

Education

The University of Auckland

2025 – Present

Bachelor of Engineering (Honours) | GPA 8.92/9

- Top in Course Award(s) – CHEMMAT 121 & ENGGEN 115, The University of Auckland

Mahurangi College, Auckland

2022 – 2024

- Excellence Endorsement in Level 3 Mathematics, Chemistry, and Geography
- Dux Award 2024

Technical Skills

- **Design:** Fusion 360, SolidWorks, & Ansys (*Basic FEA*)
- **Programming:** C/C++, Arduino, & MATLAB
- **Hardware:** ESP32, Arduino, Embedded systems, & PCB design
- **Manufacturing:** Soldering, 3D printing, Laser Cutting, Machining, & Composite Manufacturing, FMEA
- **Soft Skills:** Teamwork, adaptability, communication, & time management
- **Languages:** Languages – Japanese & English

Projects

Electric Go-Kart Composite Chassis

March 2024 – November 2024

Technologies: Fusion, Laser Cutting, Composite Manufacturing

- Contributed to the design and manufacture of a **fully carbon fiber chassis** as part of an EVelocity competition team.
- Exposure to composite manufacturing processes including **wet laminating, vacuum bagging, mould fabrication**.
- Produced a lightweight and structurally robust frame, capable of supporting a full driver load of over 80kg.

Custom Shortcut Controller

January 2026 – May 2026

Technologies: Fusion, Arduino IDE, 3D Printing, ESP32, Soldering & Electronics

- Prototyped a custom **Human Interface Device** to enable control of devices to improve digital art workflow in order to reduce hand-strain for users.
- Designed and iterated an electronics enclosure for an ergonomic, user-centric device.
- Evaluated design constraints such as tolerancing and material limitations.
- Designed hardware and firmware to enable programmable macros and device inputs.

Additional project information: github.com/jkuw137

Mini-Warman Competition

March 2026 – June 2026

Technologies: Fusion, Arduino, 3D printing, Laser Cutting, Electronics

- Collaborated on developing an **autonomous** system capable of **collecting, transporting** and **depositing** tennis balls as part of a team.
- Conducted analyses to design gear trains and arm mechanisms to improve torque and control.
- Applied an understanding **Design for Production** (DfP) principles to enable rapid design iterations and reliable assembly.

University of Auckland Rocketry

March 2026 – Present

Technologies: Fusion, OpenRocket, Laser Cutting, Manufacturing

- Member of the University of Auckland Rocketry club, participating in collaborative projects and technical workshops focused on rocket design and manufacture.
- Using Software to simulate a rocket to analyse centre of pressure and gravity to predict performance.

NZ Undergraduate Space Challenge

July 2026 – Present

- Member of a multidisciplinary team designing a payload module for integration into a 3U CubeSat to be launched into Low Earth Orbit.
- Collaborating across multiple disciplines to develop practical solutions for a real-world space mission.

Work Experience

Swimming Instructor (Contractor)

2023 – 2024

- Delivered structured swimming lessons to children aged 5-14, adapting to varying ability levels.
- Monitored and tracked the progress of individual students over time.
- Developed strong foundations in soft skills such as adaptability, time management, communication, teamwork, and leadership in group environments.

Accounting Assistant (Contractor)

2020 – 2025

- Supported day-to-day technical operations at Small Business Accounting (SBA) Warkworth as a contractor including IT infrastructure and network structure setup and maintenance, MS Office 365 user support.
- Developed an automated reporting process using Excel macro functions. Improving efficiency and reducing manual workload as well as consolidating the client database.
- Assisted in bookkeeping, invoicing, and GST preparation using Xero accounting system.

Achievements

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| • Cumulative GPA of 8.92/9, The University of Auckland | 2026 |
| • Top in Course Award – CHEMMAT 121 & ENGGEN 115, The University of Auckland | 2025 |
| • Dux Award, Mahurangi College | 2024 |
| • NZQA Scholarship Award – Physics and Chemistry | 2024 |
| • Evolocity Auckland Engineering Design Award | 2024 |

Extracurriculars and Volunteer experience

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|---|----------------|
| • Executive Member, Project Wellbeing Engineering | 2025 – present |
| • Member, University of Auckland Rocketry club | 2026 – present |
| • Student Volunteer, Robogals UoA | 2026 – present |
| • Volunteer, Kaipātiki Project | 2025 – present |
| • Volunteer, The Jane Gifford Trust | 2023– 2024 |

Interests

- CAD design – Designing for personal and academic projects, developing a strong proficiency in CAD.
- Competitive Swimming – Trained and competed at a national level, developing teamwork, discipline and time management skills.
- Digital art and design
- Piano and music